

# GAURAV KUMAR

Software Engineer (Python)

gauravatwork17@gmail.com | [github.com/ohnogaurav](https://github.com/ohnogaurav) | [linkedin.com/in/gauravconnects](https://linkedin.com/in/gauravconnects)

## PROFESSIONAL SUMMARY

Software engineer with experience building backend systems, data-driven applications, and automation pipelines using Python. Skilled in designing APIs, handling large-scale data processing, and developing reliable, maintainable systems. Strong foundation in problem-solving, system design basics, and debugging. Interested in building scalable applications and efficient software systems.

## TECHNICAL SKILLS

**Programming:** Python, SQL, JavaScript (basic)  
**Backend & APIs:** FastAPI, Flask, REST API Design  
**Data & Tools:** Pandas, NumPy, Hadoop, Spark  
**Core Concepts:** Data Structures & Algorithms, OOP, System Design Basics  
**Tools & Platforms:** Git, Linux, AWS (basic), Selenium

## SELECTED PROJECTS

### Personalised Retrieval-Augmented Chatbot System

2026

*Context-aware response generation with persistent user memory*

- Built a RAG system using Gemini API + MongoDB for persistent user context
- Designed a retrieval pipeline to generate relevant, context-aware responses
- Implemented memory-aware query handling enabling consistent, personalized interactions across sessions
- Developed a real-time interface using Gradio for interactive query handling and response visualization

### EduTrack – Location-Aware Attendance System with Liveness Verification

2025

*Secure attendance system preventing proxy marking using geolocation and identity checks*

- Built an attendance system enforcing location constraints to ensure presence within authorized zones
- Implemented liveness verification mechanisms to prevent proxy attendance and identity spoofing
- Designed backend workflows for attendance logging, validation, and real-time status tracking

### Email Rewriting API with Transformer-Based Inference

2024

*Real-time text transformation using fine-tuned language models*

- Fine-tuned a transformer model on 5,000+ email samples, improving rewriting fluency by ~30%
- Built and deployed a Flask-based REST API handling 100+ daily inference requests
- Designed a modular inference pipeline for scalable request processing and response generation
- Developed a reproducible training workflow, reducing experimentation time by ~40%

### Meal Minder – Reward & Penalty-Based Food Waste Reduction System (SIH)

2024

*Behavior-driven system to minimize food wastage in hostels and mess environments*

- Designed a system that tracks user meal consumption and applies reward/penalty mechanisms to discourage food wastage
- Implemented backend logic for logging consumption patterns and calculating user-specific penalties or incentives
- Built data-driven workflows to monitor usage trends and identify high-waste behaviors

## EXPERIENCE

### Independent Software Developer

Jan 2023 – Present

- Built multiple software projects focusing on backend development, data processing, and system reliability
- Designed reusable components and modular architectures
- Focused on debugging, optimization, and improving system performance

## EDUCATION

### B.Tech, Computer Science & Engineering

2022 – Present

Lovely Professional University, Punjab

*Relevant Coursework: Machine Learning, Probability & Statistics, Linear Algebra, Natural Language Processing*